

# GIRIDHAR CHANDRASEKARAN

## Data Scientist

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## SUMMARY

*"Data Scientist with **around 7 years of experience** with exposure to modelling data from different domains and always passionate about solving complex business solution or even day to day problems with data science."*

- Very strong NLP expertise with product experience in Social media insights mining
- Marketing and Consumer insights experience primarily in Retail and E-Commerce domain
- Developed a RNN and CNN for Text classification
- Developed a Spacy and BERT for Custom Entity recognition
- Developed a Bi-LSTM and CRF for Attribute Value Pairs(AVP) recognition
- Designed and developed Chatbot using RASA to handle multi-intent with the help of DIET transformer
- Developed a context aware search engine using BM25F and BERT embedding
- Strict DevOps and MLOps practitioner following Agile methodology and scrum concept
- Familiar with microservices and containerization have used docker and docker-compose to ship out modules
- Have leveraged AWS and Azure cloud platform services
- Adept in handling data engineering and modelling workflows
- Strong SQL and Analytical SQL skills for querying relational databases and Data Warehouses
- Hands-on experience with Big Data ecosystem tools like Hadoop, Hive and PySpark)

## EXPERIENCE (7+ YEARS)

### 5. DATA SCIENTIST

Ideas2IT Technologies

Feb 2021 – Ongoing (1.5 year)    Chennai, India

#### NLP Engineering

**CompasGo - An internal operations website empowered with a chatbot and search capability**

AWS Neptune Graph Powered internal daily use platform to search products, check Inventory, check shipping status, cross-compare competitor products, read announcement and articles and place order.

- Performed NER for Attribute Value Search on Graph using BiLSTM-CNN
- Identified similar words for Alias fields using word embedding
- Worked on Developing a complete Graph Product Solution for product data modelling using AWS Neptune, RDF
- Proposed and developed a chatbot using RASA which could perform NLU based intent identification
- Used DIET Transformer, SpaCy and Regex for Custom Entity recognition.
- For Announcement, Article and Product search used BM25F and BERT based document ranking system
- Deployed as a docker service inside AWS container

## SKILLS AND TOOLS

### • Languages

Python   R   SQL   RASA  
Chatbot development   Regex   C++  
Git   Pyspark

### • Databases

AWS Neptune   Netezza   MySQL  
Hive   MongoDB   Elasticsearch

### • Development Tools – IDE

Alteryx   Anaconda Suite   Sagemaker  
VS Code   DBeaver   Google Console  
DataBricks   MS SSMS

## INTERNSHIP

Salesforce Intern

Softsquare

Jun 2013 – Nov 2013

## 4. DATA SCIENTIST

### Purchasing Power

📅 Nov 2020 – Jan 2021 (3 months) 📍 Chennai, India

### Consumer Insights

An voluntary employee benefit E-Commerce company operating in US with a need to leverage on datascience to help consolidate their decisions

- Performed RFM Analysis and RFM based customer segmentation to understand segment response to campaigns
- Added extra demographic features to understand micro cluster patterns and removed seasonality data using Deseasonalizing techniques
- Performed standard Market basket Analysis and combined micro cluster patterns to build a recommendation engine
- Developed a end to end feedback and comment evaluation and summarization using text analytics
- Identified purchase drivers using entity recognition of SpaCy and displayed them as tags
- Using the volume of positive and negative sentiments in comments created a satisfaction score

## 3. SENIOR DATA ANALYST

### Latentview Analytics

📅 Nov 2017 – Nov 2020 (3 years) 📍 Chennai, India

### 1. Product Development Expertise

**SmartInsights - An end to end Social Media Analytics platform with Trend Identification.**

A platform to identify social media trends, Score Influencers and Identifies key drivers using organic social media conversations and thus help leverage brands to understand and focus on the rapidly changing needs of global consumers.

#### • Data Engineering:

- 75 Million conversations occupying 150GB in MongoDB hosted on AWS.
- Used API handlers, XPATH and selenium for data ingestion.
- Preprocessed the data using RegEx and rule-based filters and stored it as JSON in Elasticsearch.
- Exposed the data packets as a webservice using Flask.
- Scheduled the extraction, preprocessing and systematic backup modules using Cron Jobs.

#### • Data Modelling:

- Lexical analysis was done using NLP and NLTK and identified trends using N-grams.
- Ran spam and sentiment deep learning models on the data trained using Keras and Recurrent Neural Network.
- Modelled the Purchase Drivers and Purchase Intent modules using recurrent neural network.
- Used Convolutional Neural Network(CNN) to identify the various makeups used from a picture
- Created a brand perception map using Correspondence Analysis.
- Modelled the seasonality and removed its effect using statistical time series modeling.
- Clustered the data to identify the buckets for various purchase drivers using both Hierarchical and K means summarized the final names using K Medoids.
- Automatic identification of trends was achieved using named entity recognition (NER) deep learning model perfected using a CNN-RNN combination.

## EDUCATION

Post Graduate Diploma in Data Mining

Annamalai University

📅 Apr 2017

B.E. in Electronics and Communication Engineering

Anna University

📅 Aug 2009 – June 2013

## ONLINE TRAININGS

🏆 R Programming  
Coursera

🏆 Neural Networks and Deep Learning  
Coursera

🏆 Pyspark, Deep Learning and Statistical Thinking  
DataCamp

## CERTIFICATES

Basics of Business Analytics

Loyola Institute of business administration

📅 Nov 2016

AWS Machine Learning Specialty

Amazon Web Services

📅 Aug 2022

## LANGUAGES

English ●●●●●

Tamil ●●●●●

Hindi ●●●●●

## 2. Later Projects

### Consumer Intelligence and Marketing Analytics (for a Major Retail Brand)

Using Azure databricks and secondary data from Mobile Attribution, Web Attribution platforms, social media, survey responses and foot traffic intelligence data, perform consumer analytics and thus help the client create value proposition for their various offerings.

- Periodic Marketing KPI reports
- Text Analytics on app reviews and feedbacks.
- Measured campaign effectiveness through A/B testing and checked inferences using t-tests.
- Social media analysis : products purchase drivers and sentiments using RNN (source:sprinklr)
- Identified Brand Perception using Correspondence analysis and helped position the brand positively in the consumers view.
- Performed Market Basket Analysis and time series clustering to identify interest patterns
- Survey questionnaire preparation and response analysis to identify interest patterns
- performed consumer segmentation and identified core and growth segments.(profitable zones)
- Helped source a powerBI dashboard to provide a one-stop Insights Engine by putting all metrics in the board and created scripts in pyspark which could monitor the health of metrics highlight them if critical.

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## Past Experiences

### Retail + Insurance domains

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## 2. SOFTWARE ENGINEER

### Sopra Steria

📅 Oct 2015 – Dec 2015 (3 months)

📍 Chennai, India

- Extract Transform and Load from IBM Netezza and Hbase using python
- Correction of text and image encoding issues when using online forms by using NLTK and POS constructs
- SQL Optimization : 30% reduction in query timing
- SQL Optimization : reduction of higher order joins by using Inner query
- Designing and coding database tables to store the application's data
- Create pipelines to feed data to the portfolio dashboards
- Worked on soft migration to fix decimal digit issues and field type issues in RDBMS

## 1. PROGRAMMER ANALYST

### Cognizant technology solution

📅 Nov 2013 – Oct 2015 (1 year and 11 months)

📍 Chennai, India

part of a team which built an HR management dashboard providing ISO standard people management dashboard showcasing workforce, diversity, recruitment, productivity, Costs, Skills, Compliance and Succession

- Transported dashboard data from SQL Server to Azure Analysis services
  - Built a data model of 40+ Tables in Azure Analysis Services
  - Used Alteryx to perform complex data pipelines
  - Modelled diversity using HHI-Score
  - Worked on Tableau to build some of the dashboards
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